

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) Write a C code to create a process using fork() system call.
- b) Read the user input as shell command.
- c) Use a system call exec() to execute the shell command.
- d) Use a wait() system call for waiting a parent process.
- e) Display the PIDs of parent and child processes.

```
GNU nano 7.2 p2t1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    char command[100];

    // b) Read user input as shell command
    printf("Enter a Linux command: ");
    scanf("%s", command);

    // a) Create a child process using fork()
    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        exit(1);
    }

    else if (pid == 0)
    {
        // Child Process
        printf("\nChild Process\n");

        // e) Display child and parent PIDs
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Executing command: %s\n", command);

        // c) Execute command using exec()
        execlp(command, command, (char *)NULL);

        // This executes only if exec() fails
        printf("Exec failed!\n");
        exit(1);
    }

    else
    {
        // Parent Process
        printf("\nParent Process\n");

        // e) Display parent and child PIDs
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        // d) Parent waits for child
        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices. \

```

siri@SIRI CHANDANA:~$ gcc -o p1 p1.c
siri@SIRI CHANDANA:~$ ./p1
Enter value of n: 23
Fibonacci Series upto 23 terms:
0 1 1 2 3 5 8 13 21
Sum = 54

Prime Numbers up to 23:
2 3 5 7 11 13 17 19 23
Count of prime numbers = 9
Reversed Number = 32
Factors of 23 are:
1 23
Sorted Array:
1 2 3 4 5 6 7 8 9 23
This is a prime number.
Size of int = 4 bytes, float = 4 bytes, char = 1 byte, double = 8 bytes, long int = 8 bytes, pointer = 8 bytes
File p1.c read successfully.
siri@SIRI CHANDANA:~$

```

```

Child process completed.
siri@SIRI CHANDANA:~$ gcc p2t1.c -o p2t1
siri@SIRI CHANDANA:~$ ./p2t1
Enter a Linux command: uname

Parent Process
Parent PID : 1081
Child PID : 1092

Child Process
Child PID : 1092
Parent PID : 1081
Executing command: uname
Linux

Child process completed.
siri@SIRI CHANDANA:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 16
On-line CPU(s) list:   0-15
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) i7 7400U
CPU family:             6
Model:                 186
Thread(s) per core:    2
Core(s) per socket:    2
Socket(s):              1
Stepping:               2
BogoMIPS:               5380.02
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm
hons:000:sc0c10 pp0f sse4.2:cn0d1ge0 pob0te0:0vbe popcnt tsc_deadline_0c 0mr xsave avx f16c rdrand byappervisor laier tse hypede0 lah0_lm 3bm
owrefetch ssbd ibrs ibpb stibp ibrs_0hadow0 ept vpid ept_ad fsgsbase tsc_0djust bml0 0vx2 smep bml2 0v0ed smap cde0ed 00.00s00t clfwb
ase0opt arx xgetbv0l xsaves arx_vnni vme umip waitpkg gfni vaes vpclmulqdq rdpid movdir64b fsrm md_clear serialise ibt flush_l1d arch_

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:     Microsoft
Virtualization type:    full
Caches (sum of all):
L1d:                    384 KiB (8 instances)
L1i:                    256 KiB (8 instances)
L2:                     1 MiB (4 instances)
L3:                     6 MiB (2 instances)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):     0-15
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:             Not affected
Indirect target selection: Not affected
Itlb multihit:         Not affected
L1tf:                  Not affected
Mds:                   Not affected
Meltdown:              Not affected
Mmio state data:        Mitigation: Clear Register File
Reg file data sampling: Mitigation: Enhanced IBRS
Speculative Store Bypass: Not affected
Speculative Store Bypass disabled via prctl
siri@SIRI CHANDANA:~$

```

```

Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vnscape: Not affected

```

```
siri@SIRI CHANDANA:~$ ps
```

```

  PID TTY          TIME CMD
  320 pts/0    00:00:00 bash
 1406 pts/0    00:00:00 ps

```

```
siri@SIRI CHANDANA:~$ top
```

```

top - 13:05:03 up 53 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  22 total,   1 running, 21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.6 us,  0.0 sy,  0.0 hi, 99.4 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7787.6 total,  7080.9 free,   611.2 used,   251.7 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,  7176.4 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21880	13176	9800	S	0.0	0.2	0:02.84	systemd
2	root	20	0	3180	2200	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
55	root	19	-1	66840	19628	18500	S	0.0	0.2	0:00.61	systemd-journal
112	root	20	0	25612	7056	5176	S	0.0	0.1	0:01.69	systemd-udev
119	systemd+	20	0	21344	13212	11088	S	0.0	0.2	0:00.21	systemd-resolve
120	systemd+	20	0	91036	7992	7024	S	0.0	0.1	0:00.31	systemd-timesyn
189	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
190	message+	20	0	9672	5460	4736	S	0.0	0.1	0:00.25	dbus-daemon
207	root	20	0	17988	8856	7812	S	0.0	0.1	0:00.19	systemd-logind
224	syslog	20	0	222516	5936	4552	S	0.0	0.1	0:00.12	rsyslogd
231	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.01	agetty
244	root	20	0	107036	23216	13764	S	0.0	0.3	0:00.13	unattended-upgr
313	root	20	0	3188	1104	980	S	0.0	0.0	0:00.00	sessionleader
314	root	20	0	3204	1244	1108	S	0.0	0.0	0:00.10	Relay(320)
320	siri	20	0	6080	5384	3664	S	0.0	0.1	0:00.08	bash
324	root	20	0	6704	4660	3876	S	0.0	0.1	0:00.00	login
384	satwika	20	0	20320	11584	9456	S	0.0	0.1	0:00.18	systemd
385	satwika	20	0	21168	5396	1868	S	0.0	0.0	0:00.00	(sd-pam)
411	satwika	20	0	6080	5336	3668	S	0.0	0.1	0:00.03	bash
1055	polkitd	20	0	308172	8108	7212	R	0.0	0.1	0:00.08	polkitd
1407	satwika	20	0	9344	5464	3244	R	0.0	0.1	0:00.00	top

```
siri@SIRI CHANDANA:~$
```